ASSIGNMENT - 1

EXPLORATORY DATA ANALYSIS OF DATA SCIENCE SALARIES DATASET USING R

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1. Title

Exploratory Data Analysis (EDA) of Data Science Salaries Dataset Using R Programming

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2. Introduction

Exploratory Data Analysis (EDA) is the process of examining a dataset to understand its structure, identify patterns, detect anomalies, and summarize its main characteristics using statistical methods and visualizations.

In this assignment, the **Data Science Salaries Dataset** is analyzed using **R Programming**. The dataset contains information about employees working in data science-related roles, including experience level, employment type, salary, company location, company size, remote work ratio, and work year.

The analysis includes data cleaning, handling missing values, removing duplicates, metadata analysis, correlation analysis, visualization, and business insights.

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3. Objectives

The objectives of this project are:

- Import the dataset into R.
- Study the structure of the dataset.
- Handle missing values.
- Remove duplicate records.
- Remove unnecessary columns.
- Check data authenticity.
- Perform string manipulation.
- Perform date and time operations.
- Analyze categorical variables.

- Analyze numerical variables.
- Find correlations among numerical variables.

4.Dataset Description

Dataset Name:
Data Science Salaries Dataset (ds_salaries.csv)

The dataset contains salary information of data science professionals.

Important Attributes

Column	Description
work_year	Year of employment
experience_level	Employee experience level
employment_type	Full-time, Part-time etc.
job_title	Employee designation
salary	Salary amount
salary_currency	Salary currency
salary_in_usd	Salary converted to USD
employee_residence	Employee residence

remote_ratio	Percentage of remote work
company_location	Company country
company_size	Company size

5. Software Requirements

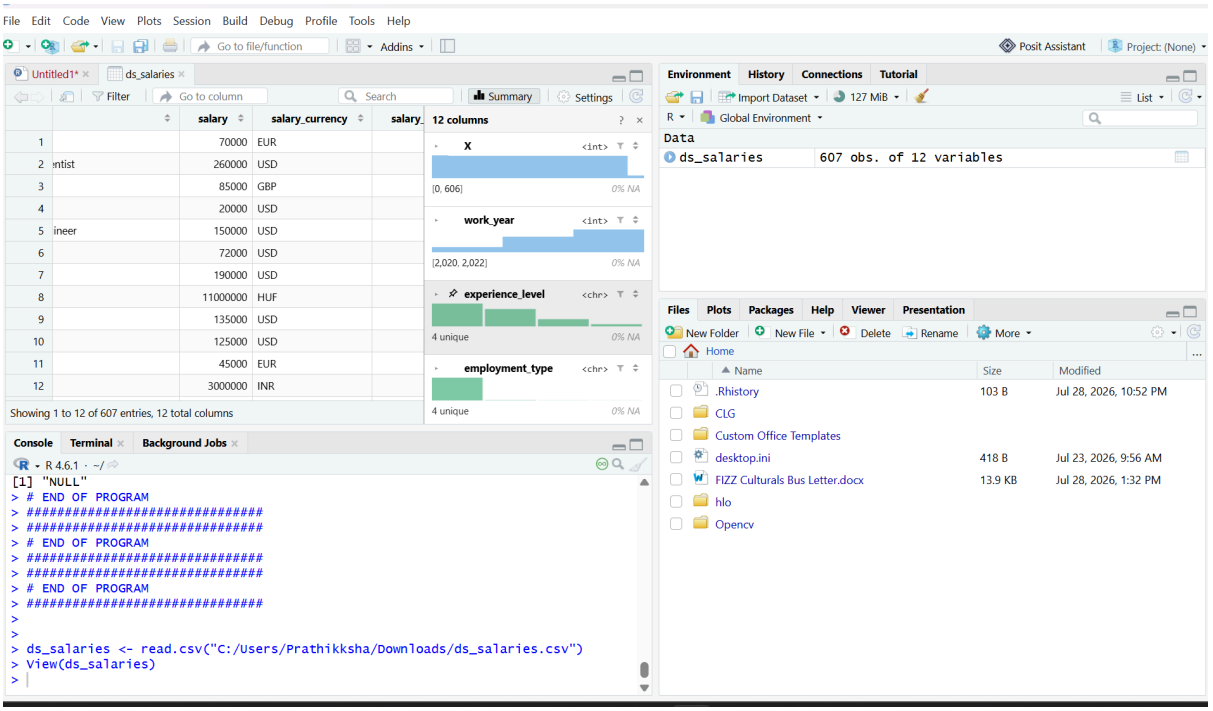
- R Programming
- RStudio
- corrplot Package

6. Methodology

Step 1 – Import Dataset

The dataset is imported using the **read.csv()** function.

```
ds_salaries <- read.csv("C:/Users/Prathikksha/Downloads/ds_salaries.csv")
```



Step 2 – Display Dataset

The first and last few records are displayed.

head(ds_salaries)

tail(ds_salaries)

```
     work_year experience_level employment_type  job_title  salary
602  601       2022             EN               FT  Data Analyst   52000
603  602       2022             SE               FT Data Engineer 154000
604  603       2022             SE               FT Data Engineer 126000
605  604       2022             SE               FT  Data Analyst 129000
606  605       2022             SE               FT  Data Analyst 150000
607  606       2022             MI               FT  AI Scientist 200000
     salary_currency salary_in_usd employee_residence remote_ratio
602              USD         52000                 CA            0
603              USD        154000                 US          100
604              USD        126000                 US          100
605              USD        129000                 US            0
606              USD        150000                 US          100
607              USD        200000                 IN          100
```

Step 3 – Dataset Information

Dataset dimensions, structure and summary statistics are obtained.

dim(ds_salaries)

str(ds_salaries)

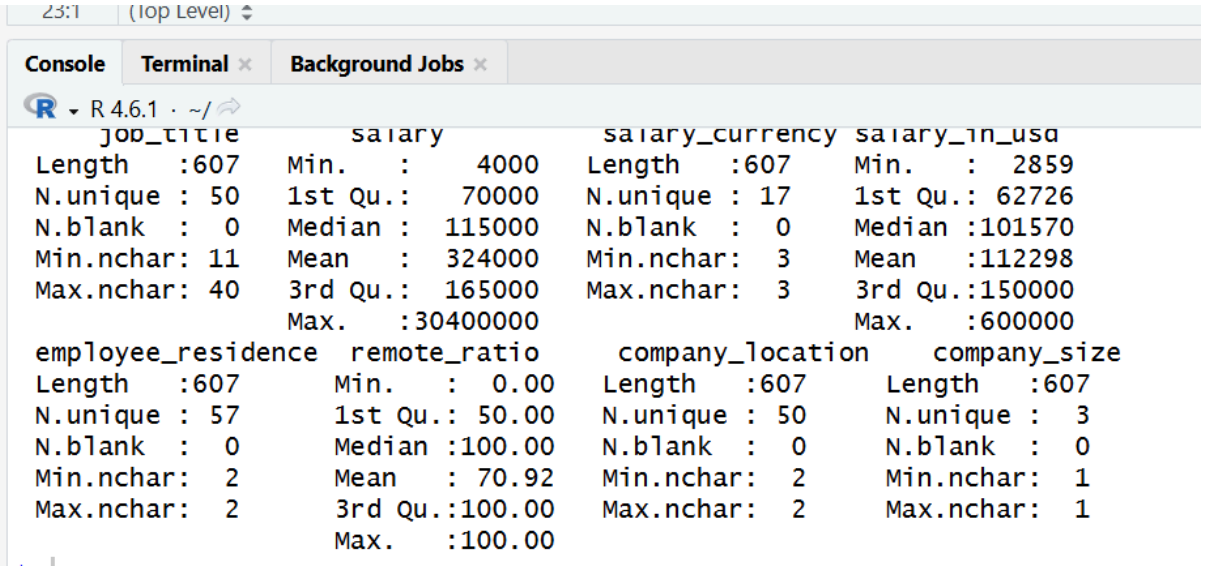
summary(ds_salaries)

Step 4 – Metadata

Metadata includes column names and data types.

colnames(ds_salaries)

sapply(ds_salaries,class)



Step 5 – Missing Values

Missing values are identified.

colSums(is.na(ds_salaries))

Rows containing missing values are removed.

ds_salaries <- na.omit(ds_salaries)

```
>
>
> colSums(is.na(ds_salaries))
                  X           work_year   experience_level    employment_type
                  0                   0                  0                  0
          job_title              salary    salary_currency      salary_in_usd
                  0                   0                  0                  0
employee_residence        remote_ratio   company_location       company_size
                  0                   0                  0                  0
```

Step 6 – Duplicate Records

Duplicate records are identified.

sum(duplicated(ds_salaries))

Duplicates are removed.

ds_salaries <- ds_salaries[!duplicated(ds_salaries),]

Step 7 – Remove Unwanted Columns

The salary_currency column is removed because salary_in_usd is already available.

ds_salaries <- subset(ds_salaries,

select=-salary_currency)

Step 8 – Data Authenticity

Unique values are checked.

unique(ds_salaries$experience_level)

unique(ds_salaries$employment_type)

unique(ds_salaries$company_size)

Invalid salary values are checked.

subset(ds_salaries,

salary_in_usd<=0)

Step 9 – String Functions

String operations performed include

- Uppercase
- Lowercase
- Search
- Replace

toupper(ds_salaries$job_title)

tolower(ds_salaries$job_title)

grep("Data Scientist",

ds_salaries$job_title,

value=TRUE)

gsub("Data","DS",

ds_salaries$job_title)

Step 10 – Date and Time

Date functions are demonstrated.

Sys.Date()

Sys.time()

Extract Year

format(Sys.Date(),"%Y")

Extract Month

format(Sys.Date(),"%m")

Extract Day

format(Sys.Date(),"%d")

Step 11 – Categorical Analysis

Frequency tables are generated.

table(ds_salaries$experience_level)

table(ds_salaries$employment_type)

```
table(ds_salaries$company_size)
```

Step 12 – Numerical Analysis

Statistical measures are calculated.

```
mean(ds_salaries$salary_in_usd)
```

```
median(ds_salaries$salary_in_usd)
```

```
sd(ds_salaries$salary_in_usd)
```

```
max(ds_salaries$salary_in_usd)
```

```
min(ds_salaries$salary_in_usd)
```

Step 13 – Correlation Analysis

Correlation matrix is generated.

```
numeric_data <- ds_salaries[sapply(ds_salaries,is.numeric)]
```

```
cor_matrix <- cor(numeric_data)
```

```
cor_matrix
```

Visualization

corrplot(cor_matrix)

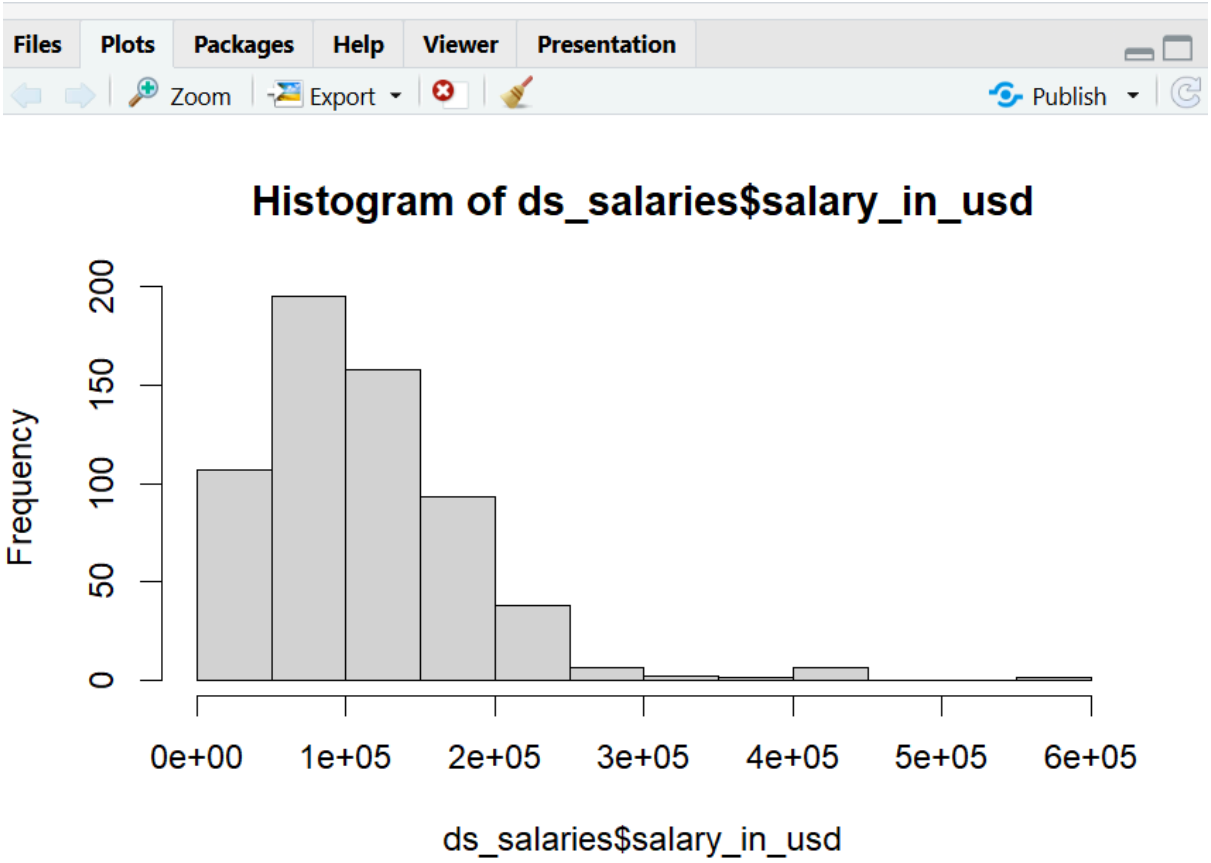
Step 14 – Data Visualization

Histogram

Shows salary distribution.

hist(ds_salaries$salary_in_usd)

Output Screenshot 14



Box Plot

Shows salary outliers.

boxplot(ds_salaries$salary_in_usd)

Bar Chart

Shows experience level distribution.

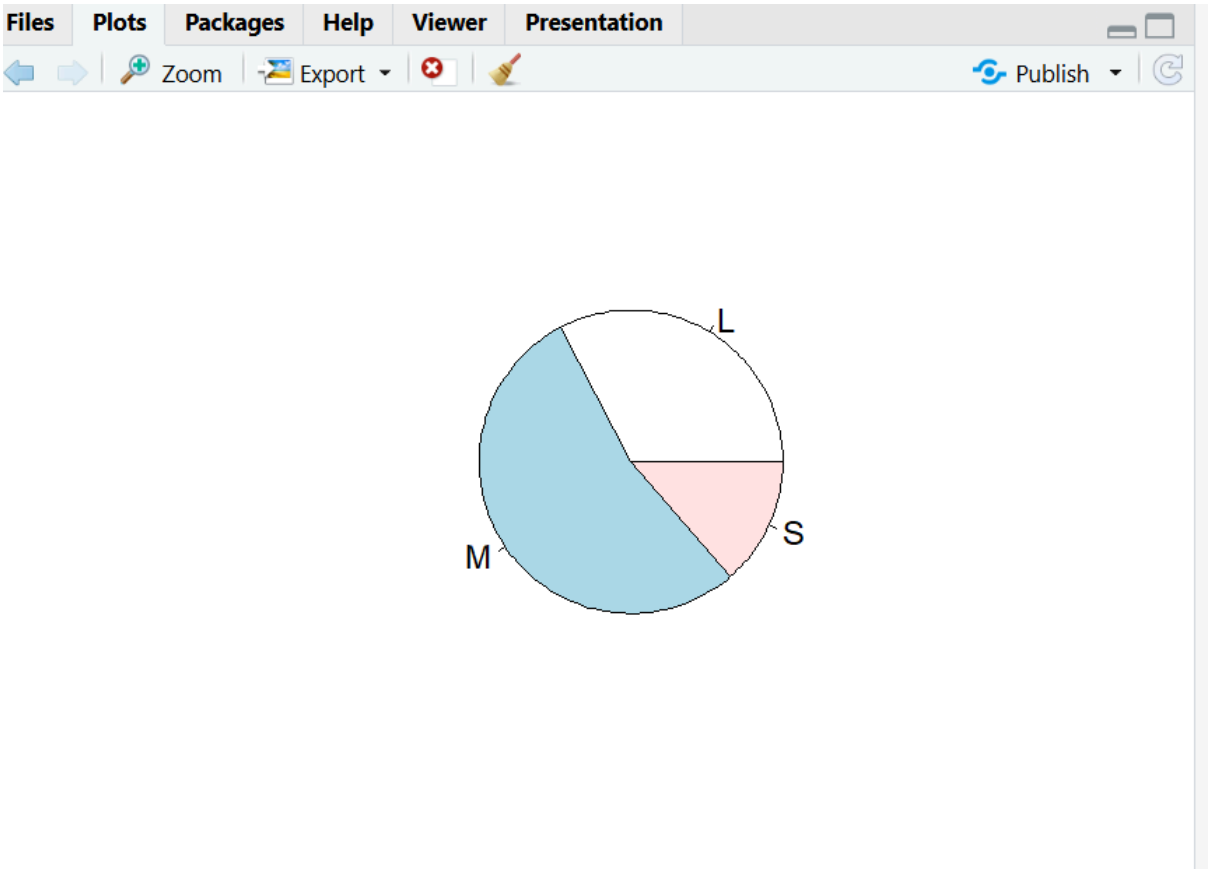
barplot(table(ds_salaries$experience_level))

Pie Chart

Shows company size distribution.

pie(table(ds_salaries$company_size))

Output Screenshot



Scatter Plot

Shows relationship between work year and salary.

```
plot(ds_salaries$work_year,

ds_salaries$salary_in_usd)
```

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7. Business Analysis

Average salary based on experience.

```
aggregate(salary_in_usd~experience_level,

data=ds_salaries,

mean)
```

Average salary by company size.

```
aggregate(salary_in_usd~company_size,

data=ds_salaries,

mean)
```

Average salary by employment type.

```
aggregate(salary_in_usd~employment_type,

data=ds_salaries,

mean)
```

Highest paying countries.

```
aggregate(salary_in_usd~company_location,
```

data=ds_salaries,

mean)

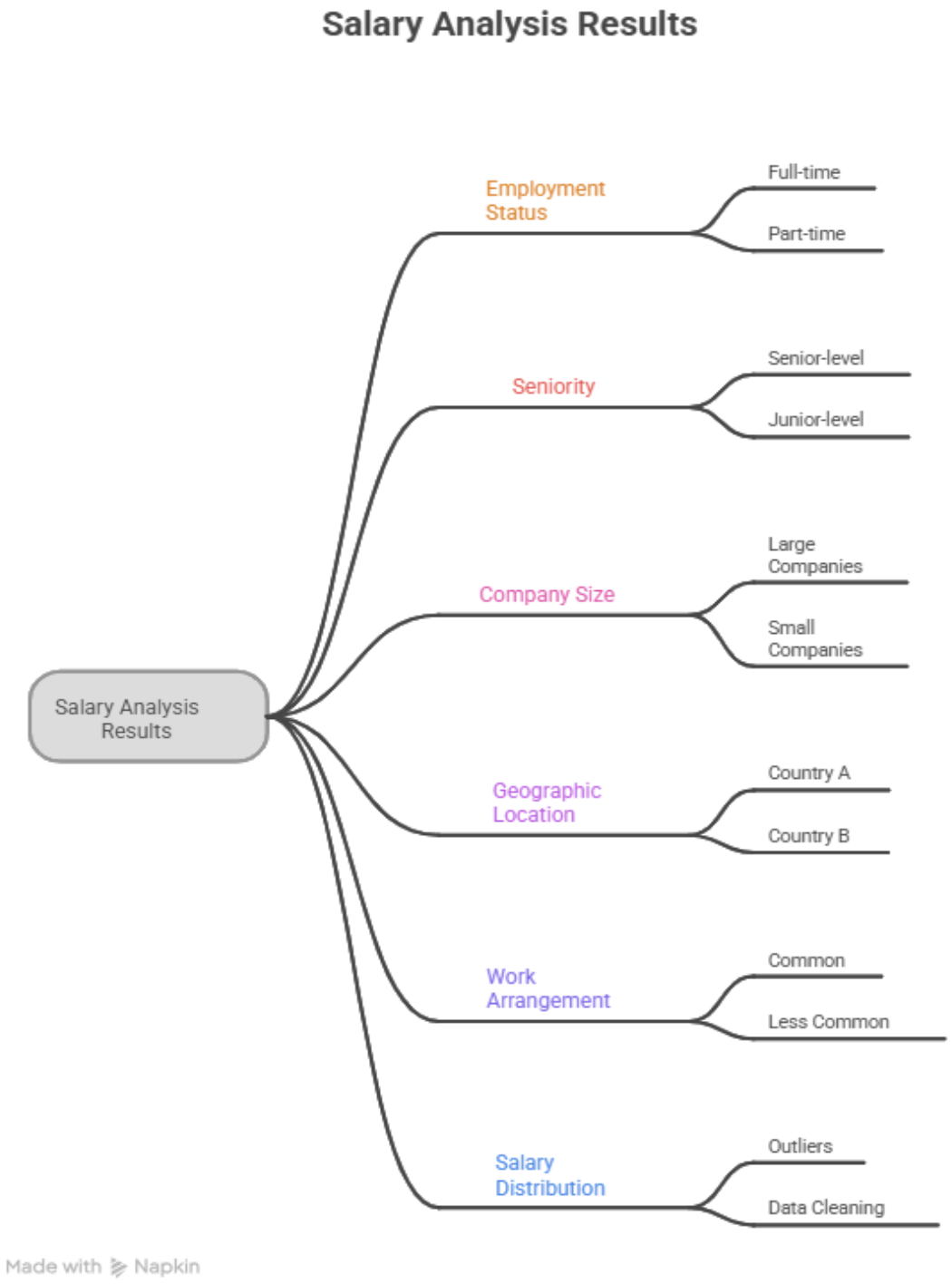
Highest paying job titles.

aggregate(salary_in_usd~job_title,

data=ds_salaries,

mean)

8. Result



9. Conclusion

The Data Science Salaries dataset was successfully analyzed using R Programming. Data preprocessing techniques such as handling missing values, removing duplicate records, checking metadata, and validating data authenticity improved the quality of the dataset. Statistical analysis and

graphical visualizations helped identify salary patterns, relationships, and trends. The study concludes that experience level, company size, employment type, and company location significantly influence salaries in the data science field.

10. References

1. R Programming Documentation
2. CRAN Package Documentation
3. Data Science Salaries Dataset
4. RStudio Documentation